

ParcelPro Manual Testing Document

1. Introduction

This document records manual testing carried out for the ParcelPro web-based parcel management system. The purpose of testing was to confirm that the system met the functional requirements described in the project report and behaved correctly during normal and invalid use.

Manual testing was used because many parts of the system involve full user workflows, including multi-step form completion, Stripe redirection, role-based access, and staff/admin interaction.

The full set of test cases is provided in this document, while the main report includes representative examples and discussion of the testing approach.

2. Test Environment

Item	Details
Application	ParcelPro
Backend	PHP
Database	PostgreSQL
Server	Apache on Debian VM
Browser	Safari / Chrome

Payment Mode	Stripe test mode
SMS Mode	Twilio integration/logged SMS
Test URL	https://debproj-ard38.teachvm.aber.ac.uk/index.php

3. Test Data

Data Type	Example
Admin account	admin@test.com
Staff account	staff@test.com
Customer account	user@test.com
Stripe test card	4242 4242 4242 4242
Valid postcodes	SY23 1AA, YO222QQ, YO16 6NN, YO13 3NN, WW134LL
Invalid postcode example	AB12 3CD if not in operating areas
Invalid phone example	12345
Valid phone example	UK mobile format

4. Manual Test Cases

Test ID	Requirement	Description	Steps	Inputs	Expected Result	Actual Result	Pass/Fail
TC1	FR1	Register new customer account	Open login modal, enter unused email, complete registration form	Name, email, UK phone, strong password	Account is created and user is logged in	Account created successfully	Pass
TC2	FR1 / FR11	Reject weak password	Attempt registration with weak password	test123	Registration blocked with validation message	Password rejected	Pass
TC3	FR2 / FR3	Complete guest parcel booking	Open send page, complete all required fields as guest	Valid sender, recipient, parcel, phone	User reaches summary/payment page	Summary page displayed	Pass
TC4	FR4	Reject postcode outside operating area	Enter valid-format postcode not in database	Unsupported postcode	User cannot continue	Continue button remains unavailable	Pass
TC5	FR4	Accept postcode in operating area	Enter postcode stored in <i>operating_areas</i>	SY23 1AA	Form accepts postcode and button becomes active	Button is active	Pass

TC6	FR3	Show same-postcode warning	Enter same sender and recipient postcode	Same postcode twice	Warning displayed but user can continue	Warning displayed	Pass
TC7	FR5	Generate estimated delivery date	Complete order with valid delivery type	Collection/dr op-off	Estimated date shown/excludes weekends	Date generated	Pass
TC8	FR6	Complete Stripe test payment	Continue to payment and use test card	4242 4242 4242 4242	Payment succeeds and confirmation page shown	Confirmation page shown	Pass
TC9	FR7	Track valid parcel reference	Enter valid reference number	Existing PP.. .reference	Tracking details displayed	Details displayed	Pass
TC10	FR7	Reject invalid tracking reference	Enter invalid reference	12345	Search rejected with validation feedback	Input rejected	Pass
TC11	FR8	Staff updates parcel status	Login as staff, open order, update status/location	Status + location	Order status and tracking history updated	Update saved	Pass
TC12	FR8 / FR11	Staff cannot cancel order	Login as staff and attempt cancellation	Cancelled status	Action unavailable	Staff prevented	Pass
TC13	FR9	Admin adds operating area	Login as admin, add postcode and city	Valid postcode/city	Area added and usable in send validation	Area added	Pass

TC14	FR9 / FR11	Admin cannot disable own admin access	Attempt to remove own admin access	Own admin account	Request rejected	Action rejected	Pass
TC15	FR10	SMS attempt is logged	Trigger order confirmation/status update	Valid order/contact phone	SMS attempt recorded in database	Log entry created	Pass / partial delivery
TC16	FR11	CSRF protection rejects invalid request	Submit sensitive action with missing/invalid token	Missing token	Request rejected	Request rejected	Pass
TC17	FR11	SQL-like input handled safely	Enter SQL-like string into registration/search field	' OR 1=1 --	Input does not break query or expose data	No SQL execution observed	Pass
TC18	FR11	Output escaping prevents rendered HTML	Enter HTML-like text in allowed field	<script>alert(1)</script>	Text displayed safely, not executed	Not executed	Pass
TC19	Account management	Change password with correct current password	Login, open security page, change password	Current + new password	Password updated	Password updated	Pass

TC20	Account management	Reject password change with wrong current password	Attempt password change with wrong current password	Wrong password	Change rejected	Rejected	Pass
TC21	FR3 / FR11	Reject invalid name input	Enter numbers/symbols in sender or recipient name field	John123	Field rejected with validation feedback	Rejected	Pass
TC22	FR3 / FR11	Reject invalid phone number	Enter invalid phone in contact field	12345	User cannot continue	Rejected	Pass
TC23	FR3	Allow optional email to be empty	Complete send form without email but with valid phone	Empty email	User can continue	Continued	Pass
TC24	FR3	Validate optional email if provided	Enter invalid email format	test@wrong	Email rejected	Rejected	Pass
TC25	FR3	Reject missing required sender fields	Leave sender name/address/city/postcode empty	Empty required fields	Section cannot be completed	Blocked	Pass

TC26	FR3	Reject missing required recipient fields	Leave recipient name/address/city/postcode empty	Empty required fields	Section cannot be completed	Blocked	Pass
TC27	FR3 / FR4	Reject invalid postcode format	Enter invalid UK postcode format	ABCDE	Postcode rejected before area check	Rejected	Pass
TC28	FR4	Reject valid-format postcode not in operating areas	Enter postcode that looks valid but is not active in DB	AB12 3CD	User cannot continue	Blocked	Pass
TC29	FR3	Reject invalid parcel weight	Enter zero or negative weight	0 / -1	Weight rejected	Rejected	Pass
TC30	FR3	Reject invalid parcel dimensions	Enter invalid length/width/height	0	Dimension rejected	Rejected	Pass
TC31	FR3	Reject invalid quantity	Enter zero quantity	0	Quantity rejected	Rejected	Pass

TC32	FR3	Reject invalid parcel value	Enter negative parcel value	-10	Value rejected	Rejected	Pass
TC33	FR3	Keep later sections locked until previous section is valid	Attempt to interact with later form section before completing current section	Incomplete contact/sender details	Later section remains disabled	Disabled	Pass
TC34	FR3	Unlock next section after valid input	Complete current section correctly	Valid contact/sender details	Next section becomes available	Unlocked	Pass
TC35	FR3	Persist form data between steps	Complete send form and go to summary, then return to send page	Valid order data	Previously entered data remains available	Data retained	Pass
TC36	FR3	Edit summary details from review page	Click edit button for phone and email/sender/recipient/parcel details	Existing order session	User returns to correct section	Correct section opened	Pass
TC37	FR6	Prevent payment until conditions are accepted	Attempt to pay without ticking required agreement boxes	Unticked boxes	Payment button remains disabled	Disabled	Pass

TC38	FR6	Allow payment after conditions are accepted	Tick both agreement boxes	Both boxes ticked	Stripe payment button becomes available	Enabled	Pass
TC39	FR6	Cancel Stripe payment	Start Stripe Checkout then cancel/return	Cancel link/back action	User returns without completed order	No order completed	Pass
TC40	FR6	Verify order only created after successful Stripe payment	Start checkout but do not complete payment	Incomplete payment	Order is not finalised	Not finalised	Pass
TC41	FR7	Validate tracking reference format	Enter badly formatted tracking number	ABC123	Tracking form rejects input	Rejected	Pass
TC42	FR7	Show tracking history for valid order	Track an order with updates	Valid reference	Current status and history shown	Displayed	Pass

TC43	FR7	Show suitable message for unknown but valid-format reference	Enter valid-looking reference not in DB	PP20260430 ABCDEF	Order not found message shown	Shown	Pass
TC44	Account management	Update profile with valid details	Change name/email/phone	Valid values	Profile updated	Updated	Pass
TC45	Account management	Reject invalid profile email	Enter invalid email in profile	bademail	Update rejected	Rejected	Pass
TC46	Account management	Save valid address	Enter valid address in My Address	Valid postcode/address/city	Address saved	Saved	Pass
TC47	Account management	Autofill sender details from saved address	Login with saved address and open send form	Saved address exists	Autofill option appears and fills sender fields	Filled	Pass
TC48	Account management	Reject password change if confirmation differs	Enter different new password confirmation	Different passwords	Password change rejected	Rejected	Pass

TC49	FR8	Staff search by reference number	Search staff dashboard using order reference	Valid reference	Matching order displayed	Displayed	Pass
TC50	FR8	Staff search by name	Search staff dashboard using customer/recipient name	Existing name	Matching orders displayed	Displayed	Pass
TC51	FR8	Staff search by postcode	Search staff dashboard using postcode	Existing postcode	Matching orders displayed	Displayed	Pass
TC52	FR8	Staff filter by active status	Select active status filter	All active	Active orders shown	Shown	Pass
TC53	FR8	Staff filter delivered/cancelled separately	Select delivered or cancelled filter	Delivered/cancelled	Matching closed orders shown	Shown	Pass
TC54	FR8	Staff update requires location	Attempt status update without location	Empty location	Update rejected	Rejected	Pass
TC55	FR8	Staff same-status update requires description	Keep current status and leave note empty	Same status, empty note	Update rejected / note requested	Rejected	Pass

TC56	FR8	Staff can add note when status remains same	Keep current status but add description	Same status + note	Tracking note saved	Saved	Pass
TC57	FR8 / FR11	Staff cannot update cancelled parcel	Login as staff and open cancelled order	Cancelled order	Update unavailable or rejected	Rejected	Pass
TC58	FR8 / FR11	Admin can cancel order	Login as admin and set status to cancelled	Cancelled status	Order marked cancelled	Cancelled	Pass
TC59	FR9	Admin search user by email/name/phone	Use Manage Users search	Existing user details	Matching user displayed	Displayed	Pass
TC60	FR9	Admin update user role	Change user role through admin dashboard	Customer to staff	Role updated	Updated	Pass
TC61	FR9 / FR11	Admin cannot disable own account	Attempt to disable current admin account	Own account	Action rejected	Rejected	Pass

TC62	FR9	Admin edit operating area	Modify postcode/city/status	Valid values	Area updated	Updated	Pass
TC63	FR9	Admin disable operating area	Disable supported postcode	Existing area	Area no longer accepted if inactive	Disabled	Pass
TC64	FR9	Admin create announcement	Add title, message, expiry, active status	Valid announcement	Announcement displayed on homepage	Displayed	Pass
TC65	FR9	Admin edit announcement	Modify announcement title/message/status	Existing announcement	Announcement updated	Updated	Pass
TC66	FR11	Direct URL access blocked for unauthenticated admin page	Enter admin dashboard URL while logged out	Admin URL	Redirected or denied	Denied	Pass
TC67	FR11	Customer blocked from staff dashboard	Login as customer and open staff URL	Staff URL	Access denied	Denied	Pass

TC68	FR11	Customer blocked from admin dashboard	Login as customer and open admin URL	Admin URL	Access denied	Denied	Pass
TC69	FR11	Staff blocked from admin dashboard	Login as staff and open admin URL	Admin URL	Access denied	Denied	Pass
TC70	FR11	CSRF missing token rejected	Submit sensitive form without token	Missing token	Request rejected	Rejected	Pass
TC71	FR11	CSRF invalid token rejected	Submit sensitive form with altered token	Invalid token	Request rejected	Rejected	Pass
TC72	FR11	SQL injection-like input handled safely	Enter SQL-like input in admin/search field	' OR 1=1 --	No data exposure or query break	Safe	Pass
TC73	FR11	XSS-like input escaped	Enter script-like text in allowed text field	<script>alert(1)</script>	Text not executed	Escaped	Pass

TC74	FR11	Disabled account cannot log in	Disable user then attempt login	Disabled user	Login rejected	Rejected	Pass
TC75	FR10	SMS logged on order confirmation	Complete paid order	Valid phone	SMS log entry created	Logged	Pass / partial delivery
TC76	FR10	SMS logged on status update	Staff/admin updates status	Existing order	SMS log entry created	Logged	Pass / partial delivery
TC77	FR10	SMS provider error recorded	Enable Twilio with restricted route	UK number from US sender	Error response stored	Error logged	Pass
TC78	FR10	System continues when SMS disabled	Set SMS enabled false and complete order	SMS disabled	Order completes and log is stored	Completed	Pass
TC79	Cookies / session	Cookie banner hides after choice	Accept/reject cookie banner	Button click	Banner no longer displayed	Hidden	Pass
TC80	Cookies / session	Core system works regardless of cookie preference	Reject optional cookies and use booking/login	Reject optional	Login/booking still works	Works	Pass

5. Summary of Results

All core functional tests passed. The main workflows for registration, parcel booking, payment, tracking, staff updates, admin management, and security controls behaved as expected. SMS delivery was partially limited by Twilio/provider restrictions, but SMS attempts were logged correctly in the database.

6. Testing Limitations

Testing was performed manually. This was suitable for end-to-end workflows but would be slower for regression testing in a larger system. Future work could include automated tests for validation functions, database operations, and access-control checks. Stripe testing used the sandbox environment, and SMS testing was limited by provider restrictions.